

Analysis of Churn in Telecom's dataset

Phase 3 Project
Eliud Kibet

Agenda

- Overview
- Business and Data Understanding
- Modeling
- Evaluation
- Recommendations
- Next Steps

Introduction

Overview

This project analyzes customer data from a select Telecom company to predict which customers are likely to leave the company (churn).

Data source:

<https://www.kaggle.com/datasets/becksddf/churn-in-telecoms-dataset?resource=download>

Project objective

This project will help the company identify customers at risk of leaving and take action to keep them

Business and Data Understanding

Telecom companies lose revenue whenever their customers cancel service and this project will help the stakeholders to undertake the following three aspects:

- a) Understand why customers leave
- b) Predict which customers will leave
- c) Find ways to reduce customer loss

Data

The dataset contains information about 3,333 customers where each row represents one customer

Key columns include:

- `state`, `area code`, `phone number`, `account length`,
`international plan`, `voice mail plan`, `customer service calls` and
`churn`

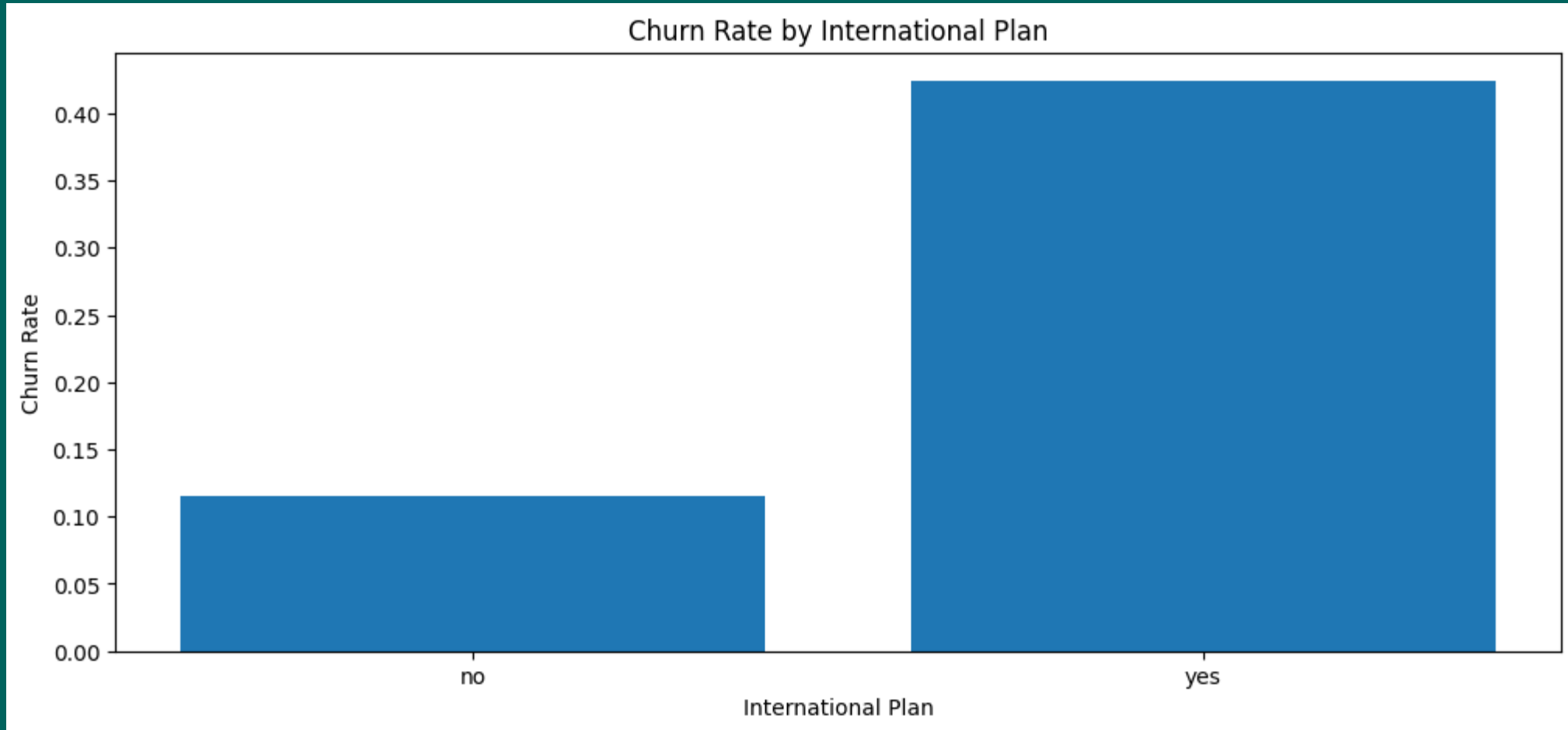


Methodology and output

Methodology

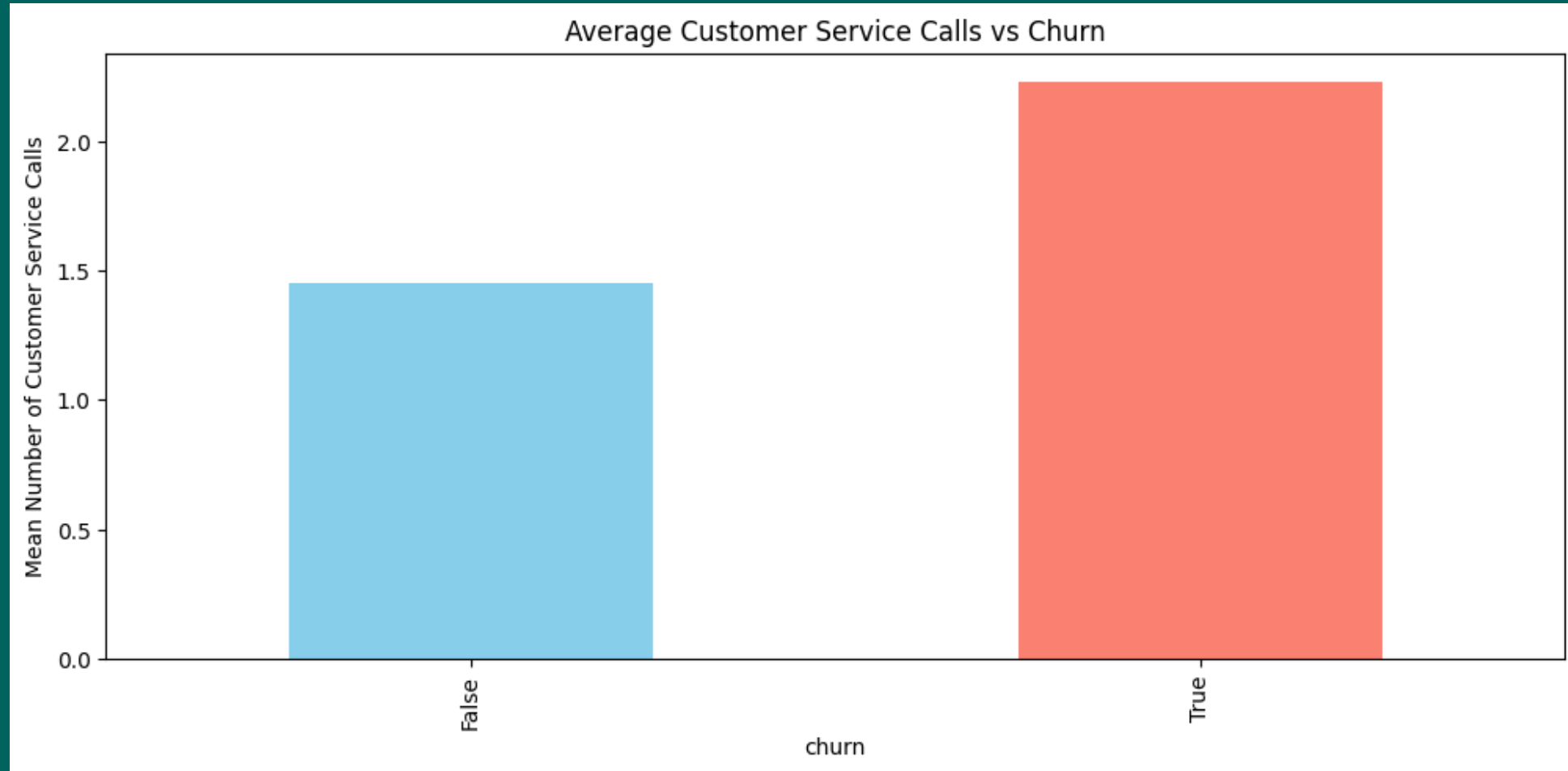
1. Exploratory Data Analysis (EDA) to Understand patterns in the data
2. There was no data cleaning done since there were no missing values. However at the point of model creation we dropped some columns to retain the most relevant features.
4. Modeling: tried two different machine learning models (a- Logistic regression and 2) Decision Tree to predict churn
5. Evaluation: Used metrics like Accuracy, Recall, Precision and F1-score
6. Recommendations: Based on model insights`

EDA output 1



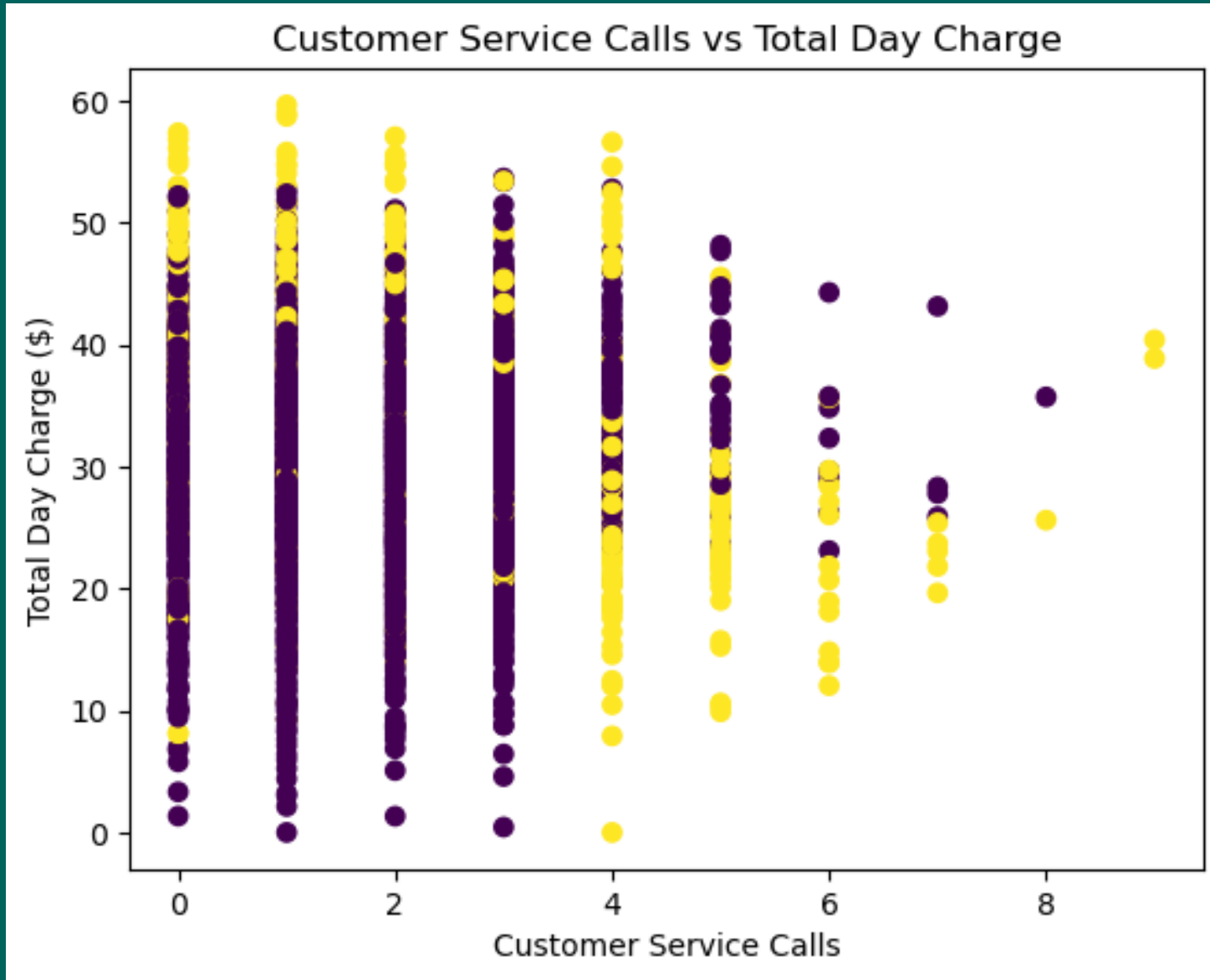
Observation: Customers who have an international plan are more likely to churn compared to those who don't.

EDA output 2



Observation: Customers with a high number of customer service calls have a much higher chance of churning.

EDA output 3



Observation: Customers with higher day charges show different churn behavior, suggesting that heavy daytime users may be more sensitive to pricing.

Modeling

Model 1: Logistic Regression

Logistic regression : 0.8620689655172413

	precision	recall	f1-score	support
0	0.88	0.98	0.92	566
1	0.63	0.22	0.32	101
accuracy			0.86	667
macro avg	0.75	0.60	0.62	667
weighted avg	0.84	0.86	0.83	667

Model 2: Decision Tree

Decision Tree : 0.9130434782608695

	precision	recall	f1-score	support
0	0.95	0.94	0.95	566
1	0.70	0.74	0.72	101
accuracy			0.91	667
macro avg	0.83	0.84	0.83	667
weighted avg	0.92	0.91	0.91	667

Evaluation

Logistic Regression	Decision Tree
1- Accuracy (86%)	1- Accuracy of 91%.
2- For customers who will Stay (0): The model is very reliable when it says someone will stay.	2- For customers who will Stay (class 0)- pthe precision of 95% and F-1 score of 95%. Recall was (94%)
3- For customers who will Churn (1): When the model predicts a customer will leave, it is correct 63% of the time (about 6 out of 10).	3- For customers who will churn (class 1) - the precision is 70% and F-1 score of 72%. Recall was 74%

Conclusions, Recommendations & next steps



Conclusion

The analysis shows that the most important factors driving churn are:

1. Number of customer service calls
2. International plan usage
3. Total day/evening charges

Recommendations

Based on the results from the models, the project recommends as follows:

- Improve customer service to reduce complaints
- Offer special discounts or better international rates to at-risk customers
- Target customers with high usage but low satisfaction
- Focus retention efforts on newer customers (low account length)

Next steps

- There is need for collection of more recent data given the changes that has happened in the sector since this sample data was collected
- There is need to include additional features such as complaints to the model to determine their influence on the model prediction
- Consider running A/B test retention offers for predicted churners

Thank You

Eliud Kibet

Github: Elky-bett

LinkedIn: elkybett

Email: elkybett@gmail.com